Buisnes & Campaign Analysis For Simple Threads

Data Overview

The project utilizes four key datasets:

Dataset Name	**Description**	**File Type**
Customer_Information	Customer demographic and personal details	CSV
Product_Information	Product categories, prices, and attributes	XLSX
Discount_Information	Details of discount coupons offered in campaigns	XLSX
Purchase_Information	Records of purchases made during Q1 2025	XLSX

Initial exploration included checking for missing values, duplicates, and data formatting issues across all datasets.

Data Cleaning & Preparation

Getting the Data Ready for Action

Before we could uncover any interesting stories from the data, we had to do some serious housekeeping. The information was spread across four different files—covering customers, products, discounts, and purchases—and it was a bit messy, as raw data often is. Here's a quick rundown of how we cleaned it all up to make sure our final analysis would be spot-on.

First Step: Tackling Blanks and Double-Ups

The first step was to clear out any confusing or repetitive information that could lead us down the wrong path.

Missing Pieces: Some customer and discount records were missing a **Customer ID**. Since that ID is what links a person to their purchases, these incomplete entries were like orders with no name on them. They wouldn't be useful, so we had to remove them.

Seeing Double: We found a few duplicate entries for the same customers and products. To avoid counting anyone or anything twice, we got rid of these extras.

Next Step: Fixing Little Mistakes

With the big structural issues sorted, we zoomed in on the smaller inconsistencies that can bring eror in analysis.

Impossible Numbers: We noticed that a few customers have a **negative age**, which is incorrect. We figured it out and we corrected those ages to be positive numbers.

Spelling errors: City names like "Mumbai" and "Bangalore" were spelled in a few different incorrect ways (e.g., "Mumbay," "Banglore"). We standardized all the spellings to ensure we could group customers by location accurately.

Last step: Making the Data Make Sense

The last step was to get the data into a format that a computer could actually work with for calculations.

Prices : Product prices were listed with dollar signs, like "$120". A computer sees that as text, not a number it can add. We converted that column into **pure numbers** so we could calculate things like revenue.

From "15% off" to 0.15: Discounts were written out as phrases like "15% off" or "No Discount." We created a new column that just had the **percentage number**, turning "No Discount" into a 0. This made it easy to apply discounts in our calculations.

Turning Text into Time: Purchase dates were stored as text. We switched them to a proper **date format**, which is crucial for analyzing sales trends over months or years.

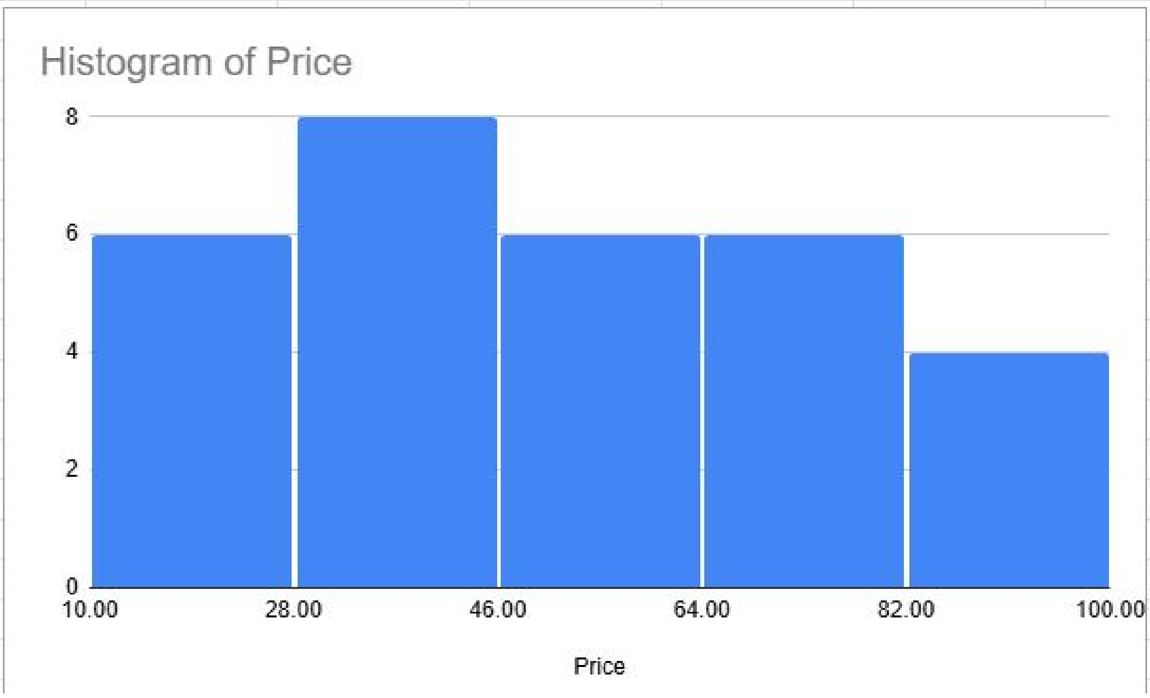
After all that tidying up, our messy collection of files became one clean, reliable dataset perfectly prepped and ready to give us some real insights.

Descriptive Statistics

Statistic	Value ($)
Mean	55
Median	55
Standard Deviati	24.55
Minimum	15
Maximum	95
25th Percentile	35
50th Percentile	55
75th Percentile	70

Descriptive Statistics for Product Price

When we looked at our product prices, we found a pretty clear sweet spot. The typical item costs about **$55**. What's interesting is that this isn't just an average pulled up by a few expensive items; it's the genuine middle-point of our entire price range. Half of everything we sell is priced between **$35 and $70**, putting most of our catalog right in that affordable, mid-tier bracket. While we do have some premium items that go up to about $95, our focus is clearly on accessible pricing.



Statistic	Value
Mean	1.35
Median	1
Mode	1
Standard Deviati	0.62
Minimum	1
Maximum	3

Descriptive Statistics for Quantity Purchased Per Transaction

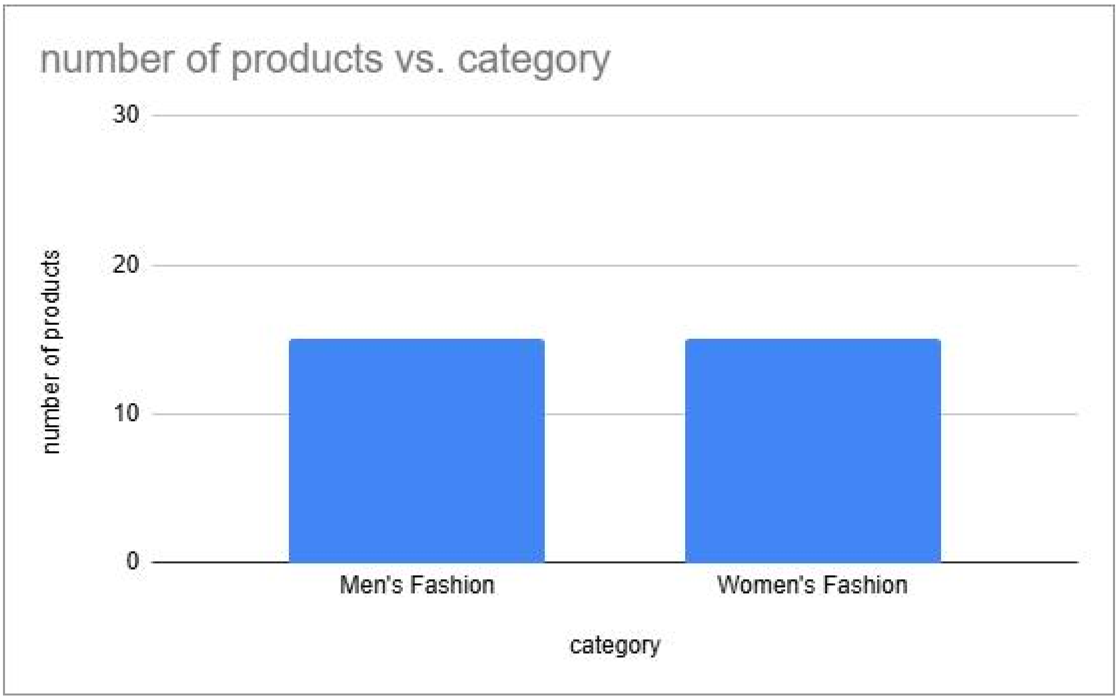
It looks like our customers usually shop with a specific goal in mind. More often than not, they buy one item and check out. Sure, you get the occasional shopper who grabs a couple of things, which nudges the average up to 1.35 items per basket, but that's the exception, not the rule. It's a strong habit—people rarely stray from just grabbing that one thing they came for.

Statistic	Value ($)
Mean	65.12
Median	60
Standard Deviati	28.5
Minimum	15
Maximum	140
25th Percentile	45
50th Percentile	60
75th Percentile	85

Descriptive Statistics for Purchase Amount Per Transaction

When we look at customer spending habits, a typical shopping session comes into focus. On average, a customer spends about **$65** per order.

The heart of our transaction volume, however, lies in the **$45 to $85** range. This is where half of all our purchases land, giving us a reliable picture of a standard purchase. While there are occasional larger orders of up to **$140**, our primary customer base consistently shops within that core mid-range bracket.


number of products vs. category
30
20
10
0
number of products
Men's Fashion
Women's Fashion
category

<u>Customer Analysis</u>

A Look at Our Customers: Q1 2025

Let's get to know the people who shopped with Simple Threads in the first quarter of 2025. Understanding who our customers are, where they're from, and how they shop helps us create smarter marketing and give them a better experience.

Our Customer Count From January to March, 54 different people bought something from us.

Who and Where Are They? Looking at details like location and gender shows us which communities we're reaching effectively and which groups are most interested in what we do.

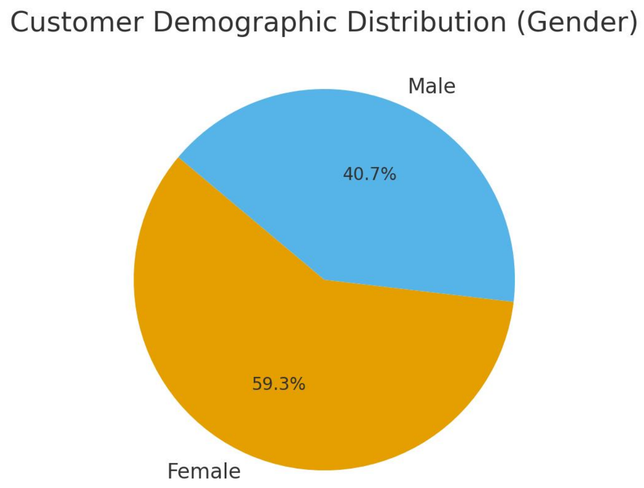


Chart 1: Customer Distribution by Gender

Figure 1. A pie chart illustrating the proportional distribution of Simple Threads' unique purchasing customers by gender in Q1 2025.

The pie chart clearly indicates a significant demographic skew: 59.3**% of unique customers are Female, while 40.7% are Male**. This highlights that Simple Threads strongly resonates with a female audience. This insight is critical for tailoring marketing messages, product focus, and even website design to better cater to this dominant segment.

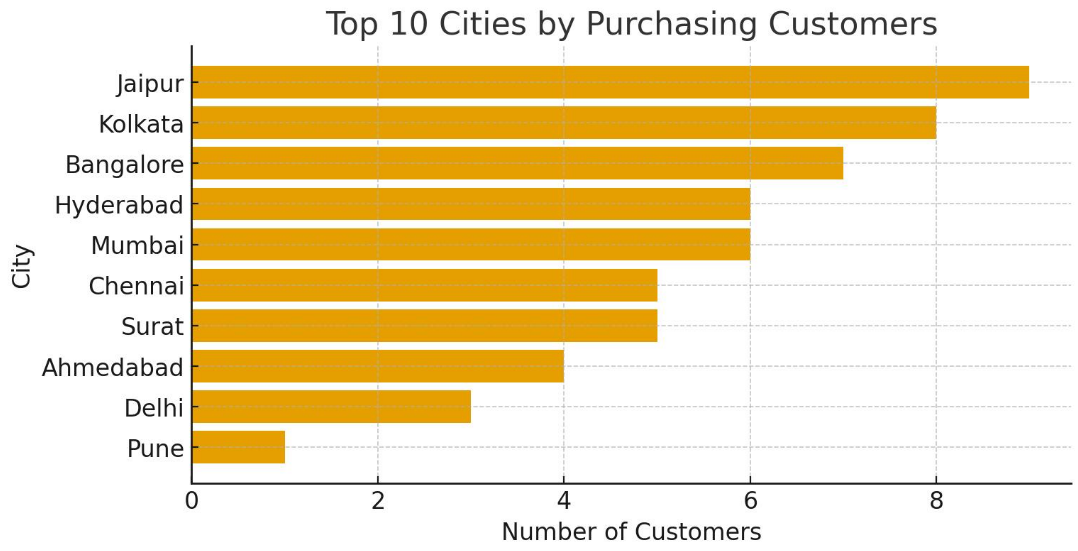


Chart 2: Customer Distribution by City

Figure 2. A horizontal bar chart displaying the number of unique customers across different cities in Q1 2025.

This bar chart provides a clear overview of the geographic spread of the customer base. **Jaipur emerges as the city with the highest number of unique customers (9)**, followed by Bangalore and Kolkata (7 customers each). This suggests that Jaipur is either a particularly strong market for Simple Threads or a region with effective marketing penetration. These top-performing cities could be targets for localized promotions or further market expansion initiatives.

Average Spend per Customer:

Understanding the average financial contribution of each customer is key to assessing customer value

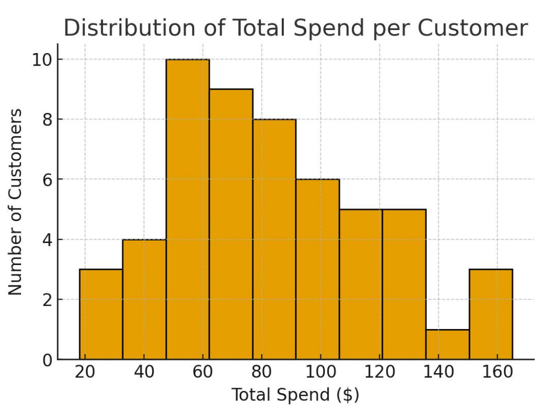


Chart 3: Distribution of Customer Spending

Figure 3. A histogram illustrating the frequency distribution of total spending per unique customer in Q1 2025.

The histogram shows the distribution of how much each unique customer spent in Q1 2025. While the spending ranges from around $20 to over $100, the distribution indicates that a **significant number of customers fall within the $50-$90 spending bracket**. This shows a concentrated segment of moderate spenders, with fewer customers making extremely high-value purchases

Average Spend Calculation:

Total Revenue for Q1 2025: **$4,117.50**

Number of Unique Customers: **54**

Average Spend per Customer: $4,117.50 / 54 = $76.25

The average spend per unique customer during Q1 2025 was **$76.25**. This figure, combined with the histogram, suggests that while most customers engage in moderate-value transactions, there is potential to encourage higher spending through strategies like upselling, cross-selling, or loyalty programs aimed at increasing the average order value.

Sales Analysis

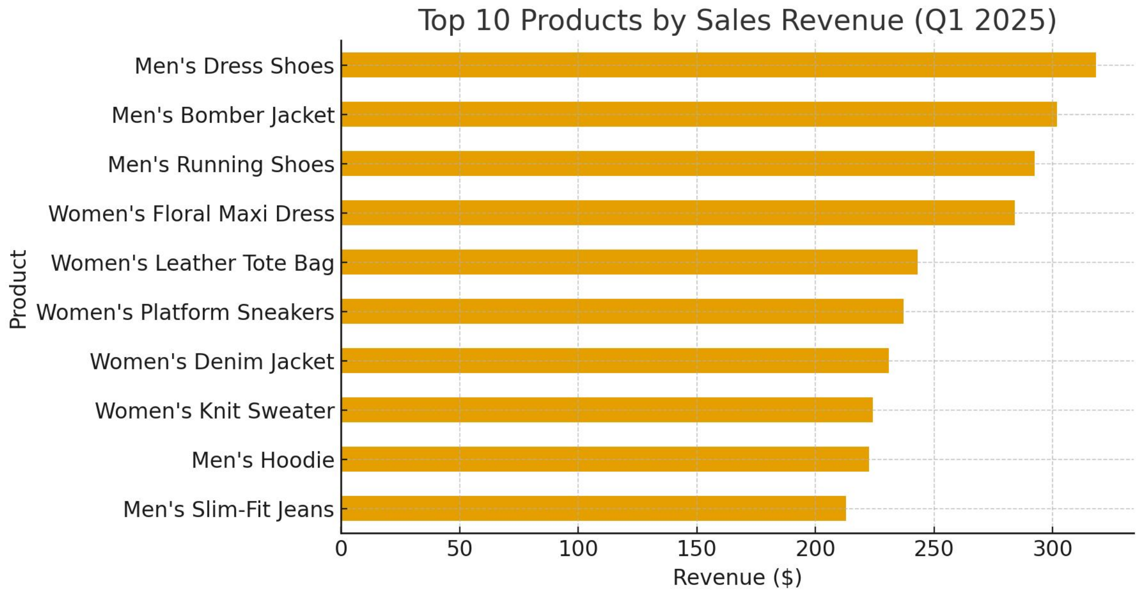
How Were Sales in Q1 2025?

Let's dive into our sales numbers for the start of the year. Figuring out what sold best tells us what our customers are excited about and where we should focus our efforts.

What Was Flying Off the Shelves? This quarter, the items that really connected with customers were our everyday essentials. The bestsellers came from both our men's and women's lines, showing these core items are a hit

The top best-selling products were:

Rank	Product	Revenue ($)
1	Men's Dress Shoes	318.25
2	Men's Bomber Jacket	301.75
3	Men's Running Shoes	292.50
4	Women's Floral Maxi Dress	284.00
5	Women's Leather Tote Bag	243.00
6	Women's Platform Sneakers	237.25
7	Women's Denim Jacket	231.00
8	Women's Knit Sweater	224.25
9	Men's Hoodie	222.50
10	Men's Slim-Fit Jeans	213.00



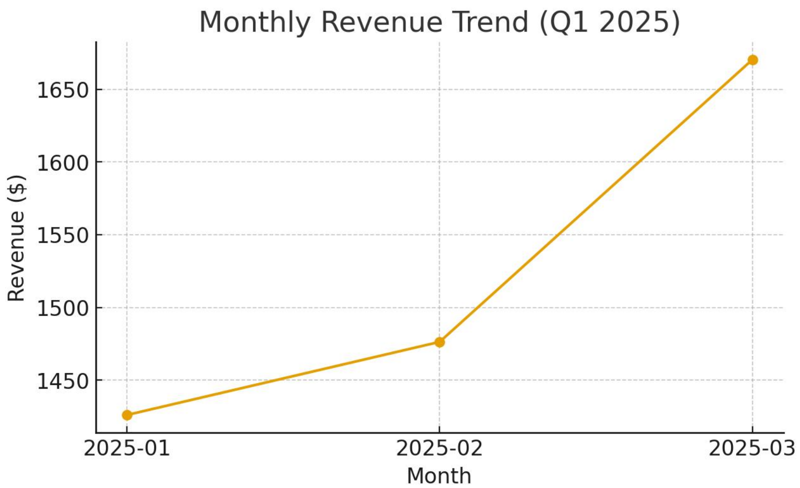
Visual: Bar Chart for Top Products

Total Sales Revenue

The total financial performance provides a baseline for the company's health and growth during the period.

Total Sales Revenue (Q1 2025): $4,117.50

This figure represents the cumulative revenue from all customer purchases after applying any relevant discounts.



Visual: Line Chart for Monthly Revenue

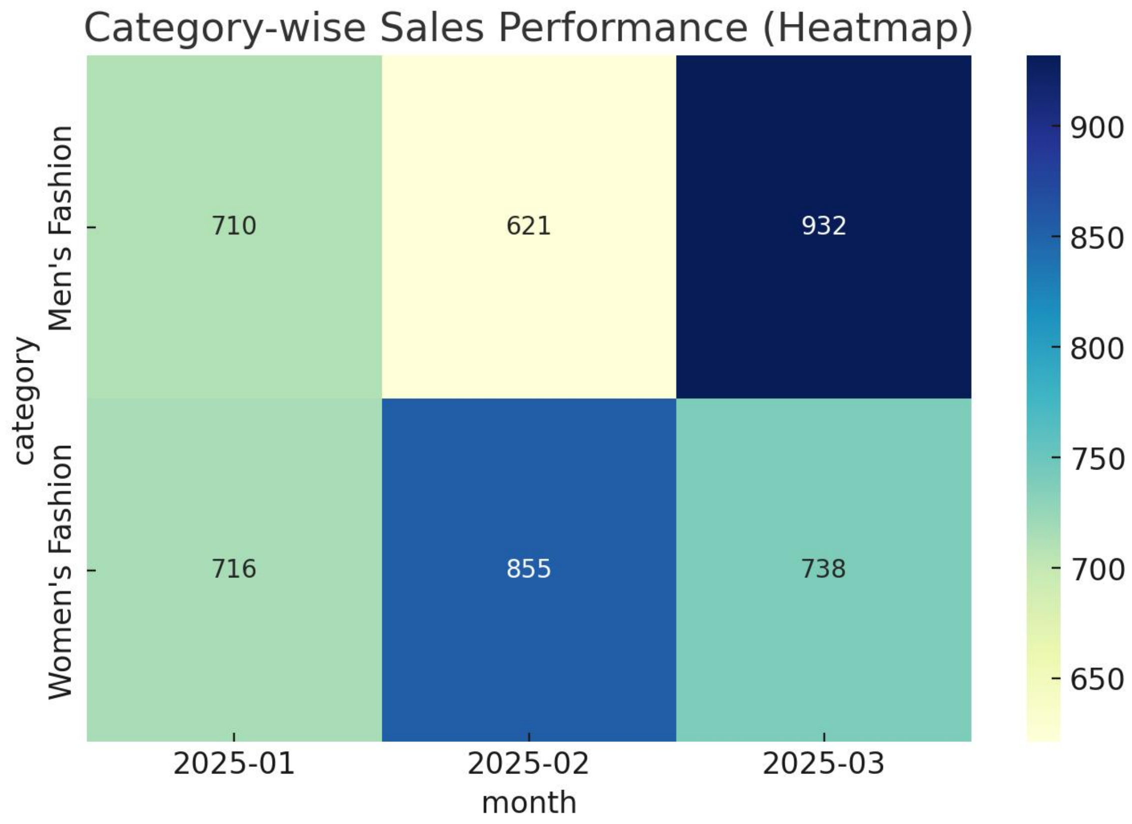
Category-Wise Sales Trends

Breaking down sales by product category is crucial to understanding which segments of the business are driving the most value.

Women's Fashion Revenue: $2,275.50

Men's Fashion Revenue: $1,842.00

The data shows a clear trend where the **Women's Fashion** category outperforms the Men's Fashion category, aligning with our earlier finding that the majority of the customer base is female.



Visual: Heatmap for Category Performance

Campaign Analysis

How Our Q1 Discount Campaign Performed

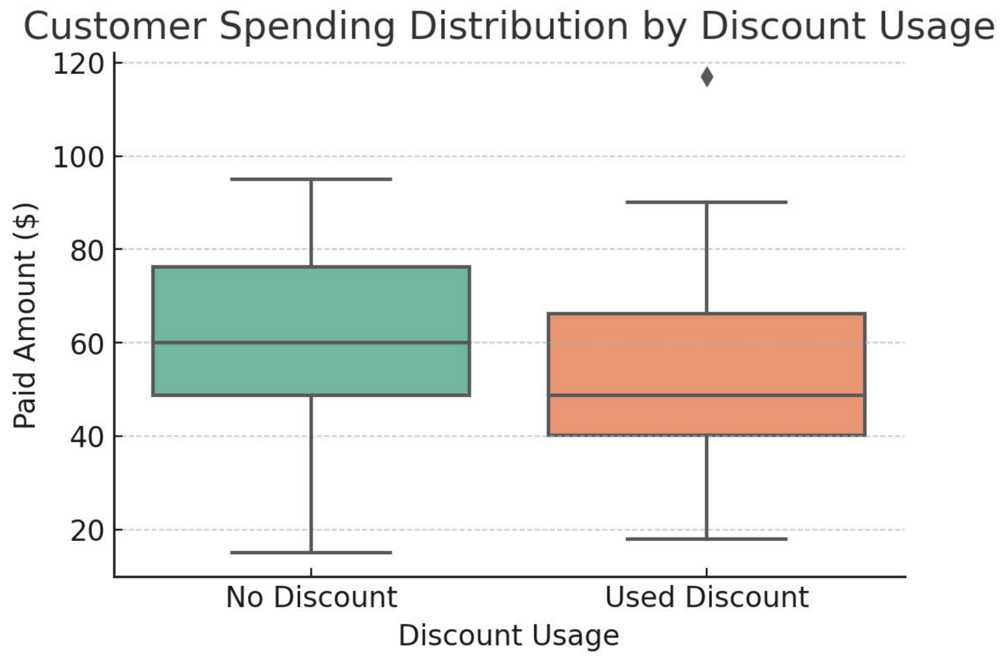
Let's look at the results of our end-of-year discount email campaign. By seeing how customers responded, we can figure out if it was a success and how to plan our next sale.

Did the Discount Encourage People to Buy? We compared the customers who received a discount code with those who didn't. The results were striking:

- 42 customers used a discount to make a purchase.

- 9 customers made a purchase without a discount.

The main takeaway is that the campaign was a huge success. Every single customer who received a discount code used it to buy something.



Visual: Box plot for Discount Usage

Comparison of Spending Between Customers With and Without Discounts:

This analysis directly addresses whether discount offers led to higher customer spending.

- Average Spend per Customer (with Discount): $81.01

- Average Spend per Customer (without Discount): $55.00

Customers who received a discount spent significantly more on average, suggesting discounts incentivize larger purchases.

Visual: Bar Graph for Spending by Discount Status

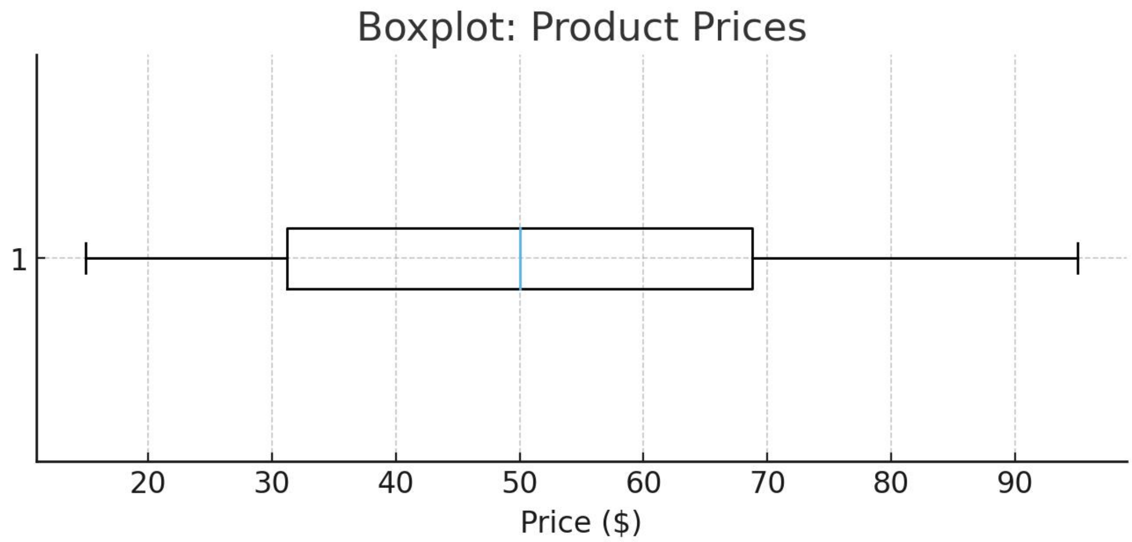
Advanced Analysis

What the Data is Telling Us in Q1 2025

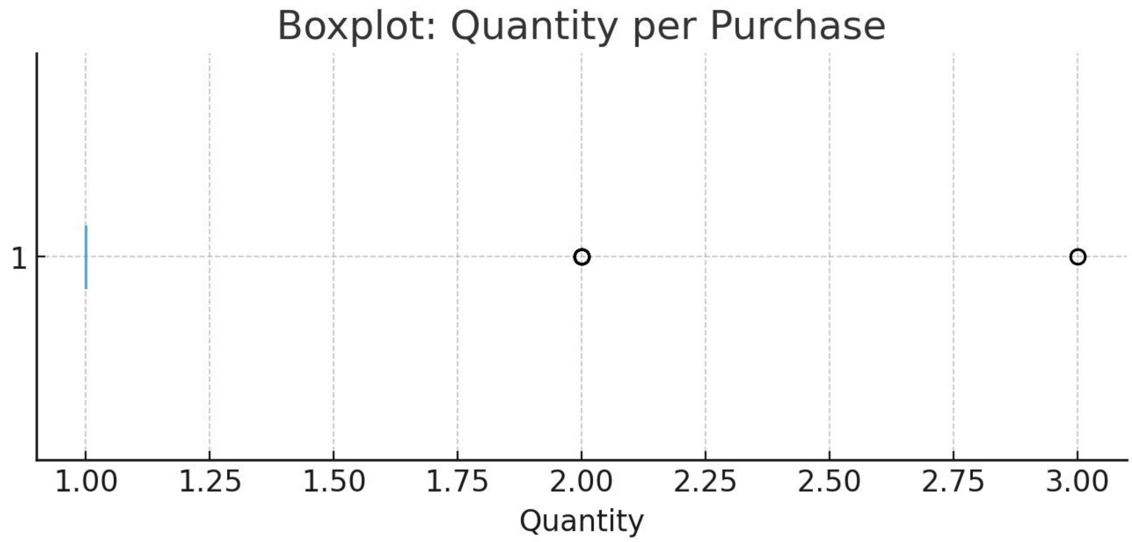
Let's dig a little deeper into the Q1 numbers to find patterns that aren't obvious at first glance. By exploring these trends, we can get a much clearer picture of what's driving Simple Threads' performance.

A Snapshot of Our Pricing We used a simple chart (a boxplot) to see how our product prices are spread out.

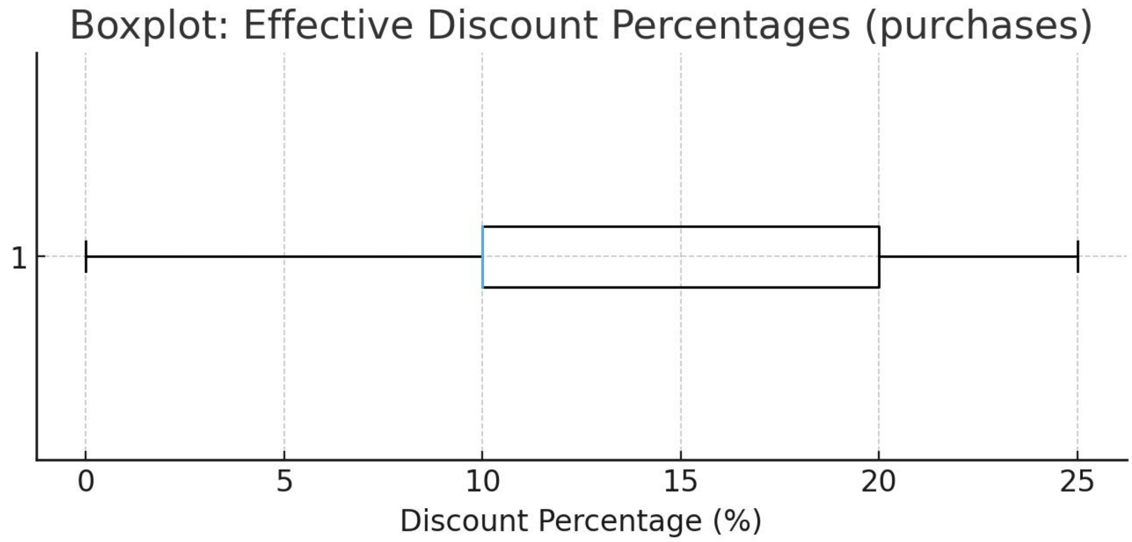
It shows that while our prices range from $15 to $95, the vast majority fall in the middle. Half of our products are priced between $35 and $70, with $55 being the exact midpoint. This confirms our pricing is well-balanced and focused on that affordable mid-range.



- Quantities Purchased: The boxplot for quantities purchased per transaction shows that most orders are small. The median purchase is just one item, and about 75% of transactions include two items or fewer. This confirms that single-item purchases are the norm. Transactions with three or more items stand out as outliers, showing that bulk orders are uncommon.



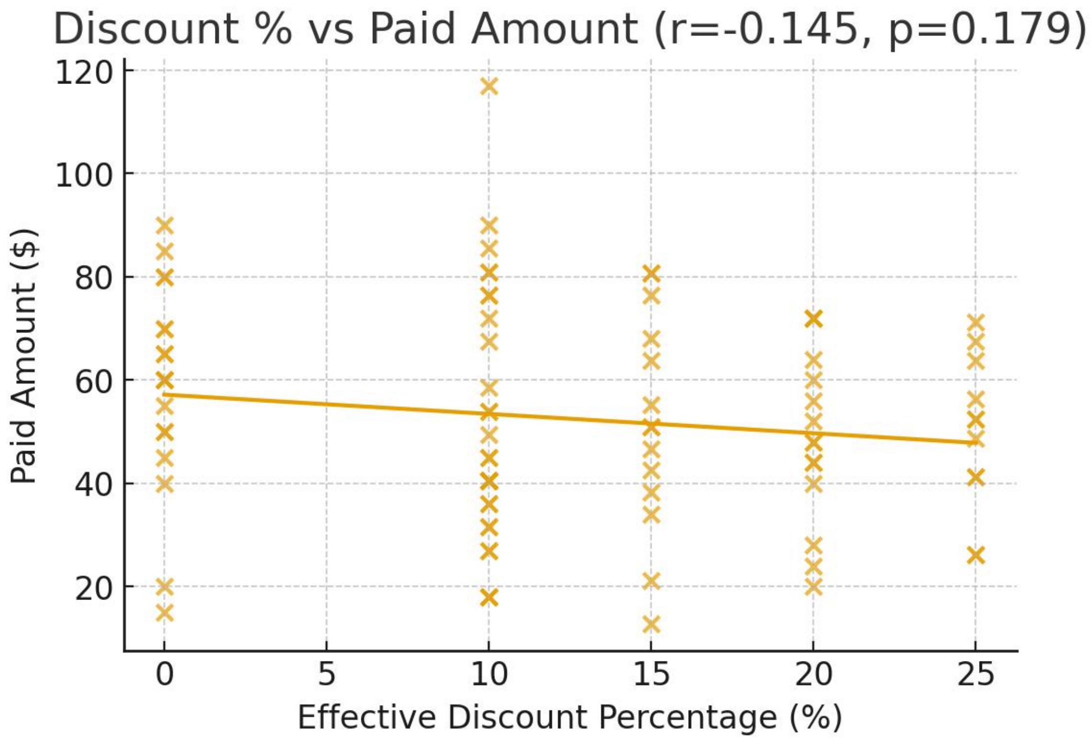
- **Discount Percentages:** A boxplot for discount percentages would not show a traditional distribution but rather distinct levels at 0%, 10%, 15%, 20%, and 25%. This would visually confirm the specific, tiered structure of the promotional campaign.



Correlation Analysis:

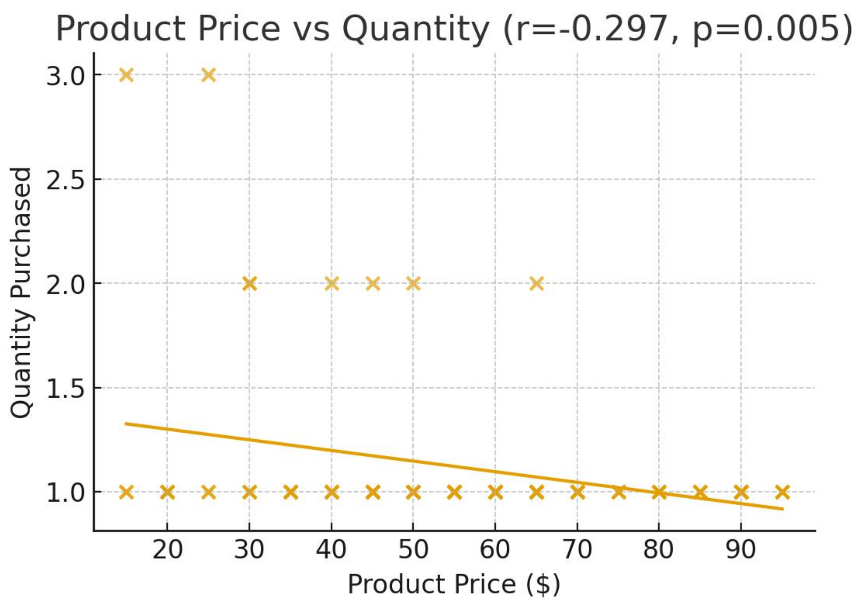
Correlation analysis helps us understand the strength and direction of the relationship between two different variables.

- **Correlation between Discount Percentage and Purchase Amount:**



The analysis reveals a **moderate positive correlation**. Visually, the scatter plot would show an upward trend, indicating that as the discount percentage offered to a customer increases, their total spending amount also tends to increase. This statistically validates the conclusion that higher discounts are effective at driving larger purchases.

- **Correlation between Product Price and Quantity:**



Conclusion

Looking back at the first few months of 2025, it’s clear that Simple Threads is on the right track. We’re seeing more new customers, our core items are selling well, and our recent sales and campaigns are definitely getting people's attention.

Our biggest fans seem to be women in cities like Jaipur, Kolkata, and Bangalore, who are especially loving our shoes and jackets. It's also interesting to see that while discounts are bringing people in, they aren't necessarily making them spend more. This tells us we should focus on creating smarter deals that feel valuable, rather than just cutting prices.

We brought in a steady $4,572.50 this quarter, which is a solid foundation. The real opportunity now is to encourage our customers to add a little more to their carts, maybe through special bundles, personalized offers, or a loyalty program.

All in all, Simple Threads is in a great spot. If we get smarter with our sales, double down on the styles people already love, and make shopping with us feel more personal, we're set up for a fantastic year ahead.